Microtubules	Absent but has homologues	Present
Endoplasmic reticulum	Absent	Present
Mitochondria	Absent	Present
Cytoskeleton	May be absent	Present
DNA.	Multiple proteins fold and condense DNA which is organized into conformations like supercoiled or wound around tetramers of proteins called HU proteins.	Eukaryotes wrap their DNA around proteins called histones.
Ribosome	Smaller	Larger
Golgi apparatus.	Absent	Present
Flagella	Submicroscopic in size and composed of only one fiber	Microscopic in size; membrane bound; usually arranged as nine doublets surrounding two singlets.
Cell wall	Usually chemically complex	Usually present in plant cells and fungi (chemically simpler)

1.3.Shape and structure of cell

Cell is a simple sphere, well-defined structure, with extended process for example neuron or epithelial cells that have distinct apical and basolateral surfaces carrying out distinct functions. The abilities of cells to take such different forms depends on the following:

- Separation of a membrane called plasma membrane from the external environment which control import and export
- Cellular components that are constructed by food sources
- Genetic information and gene expression

1.3.1.Cell and solute levels

For unicellular organisms, homeostasis is necessary because the exterior environment may be subjected to significant fluctuations. For multicellular organisms, it enables individual cells to maintain internal environments that are distinct from that of the extracellular fluid.

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